U.S.N.										

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

January / February 2026 Semester End Main Examinations

Programme: B.E.
Branch: Computer Science and Engineering
Course Code: 23CS4ESCRP
Course: Cryptography

Semester: IV
Duration: 3 hrs.
Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	CO	PO	Marks
1	a)	Find the solution for the following sets of linear equations: $2x + 3y \equiv 5 \pmod 8$ $x + 6y \equiv 3 \pmod 8$	CO1	PO1	5
	b)	Encrypt the message “the house is being sold tonight” using the following ciphers. Ignore the space between words. Decrypt the message to get the plaintext i. Vigenere cipher with key: “dollars” ii. Autokey cipher with key = 7 iii. Playfair cipher with key as shown below Secret Key = $\begin{array}{\|c\|c\|c\|c\|c\|} \hline L & G & D & B & A \\ \hline Q & M & H & E & C \\ \hline U & R & N & I/J & F \\ \hline X & V & S & O & K \\ \hline Z & Y & W & T & P \\ \hline \end{array}$	CO1	PO1	10
	c)	Use a Hill cipher to encipher the message “We live in an insecure world”. Use the following key: $K= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$	CO1	PO1	5
		OR			
2	a)	Find the multiplicative inverse of each of the following integers in Z_{180} using the extended Euclidean algorithm. i. 132 ii. 24	CO1	PO1	6

	b)	Use a brute-force attack to decipher the following message. Assume that you know it is an affine cipher and that the plaintext “ab” is enciphered to “GL”. **XPALASXYFGFUKPXUSOGEUTKCDGFXANMGNVS**	*CO1*	*PO1*	**7**
	c)	Apply the extended Euclidean algorithm to find the inverse of $(x^4 + x^3 + 1)$ in $GF(2^5)$ using the irreducible polynomial $(x^5 + x^2 + 1)$	*CO1*	*PO1*	**7**
		UNIT - II			
3	a)	AES defines three different cipher-key sizes (128, 192, and 256) but DES defines only one cipher-key size (56). Contrast the advantages and disadvantages of AES over DES with respect to this difference.	*CO2*	*PO2*	**6**
	b)	Demonstrate with a neat diagram the key generation process in DES.	*CO1*	*PO1*	**6**
	c)	Using a plaintext block of all 0s and a 56-bit key of all 0s, prove the key-complement weakness assuming that DES is made only of one round.	*CO2*	*PO2*	**8**
		OR			
4	a)	Can you devise a meet-in-the- middle attack for a double DES? Justify your answer and show how triple DES is better than double DES.	*CO2*	*PO2*	**5**
	b)	Demonstrate the creation of words for key size 128-bits in the key expansion process for AES.	*CO1*	*PO1*	**5**
	c)	Write the third, fourth and fifth criteria of S-boxes and: i. Use the third design criterion for S-box 3 using the following pairs of inputs. a. 000000 and 000001 b. 111111 and 111011 ii. Use the fourth design criterion for S-box 2 using the following pairs of inputs. a. 001100 and 000000 b. 110011 and 111111 iii. Use the fifth design criterion for S-box 4 using the following pairs of inputs. a. 001100 and 110000 b. 110011 and 001111	*CO2*	*PO2*	**10**
		UNIT - III			
5	a)	i. Find the value of x for the following sets of congruence using the Chinese remainder theorem. $x \equiv 4 \bmod 5$ and $x \equiv 10 \bmod 11$ ii. Using quadratic residues, Solve $x2 \equiv 36 \bmod 143$	*CO1*	*PO1*	**10**

		b)	Find the values of the following: i $\phi(32)$ ii $\phi(101)$	*CO1*	*PO1*	**5**
		c)	Find the results of the following using Euler's theorem: i 12^{-1} mod 77 ii 21^{-1} mod 95	*CO1*	*PO1*	**5**
			OR			
	6	a)	Apply Miller-Rabin Test to check whether the following numbers are prime or not. Use base value as 2 for testing and then conclude on the results. (i) 109 (ii) 271 Show all the steps clearly for each iteration.	*CO1*	*PO1*	**8**
		b)	Find the value of x for the following sets of congruence using the Chinese remainder theorem. i) $x \equiv 4 \pmod 5$, $x \equiv 10 \pmod{11}$ ii) $x \equiv 7 \pmod{13}$, $x \equiv 11 \pmod{12}$	*CO1*	*PO1*	**8**
		c)	Find the result using Fermat's little theorem for the following equation: 15^{-1} mod 17	*CO1*	*PO1*	**4**
			UNIT - IV			
	7	a)	In RSA, given p=19, q=23 and e=3, find n, $\phi(n)$ *and* d	*CO1*	*PO1*	**4**
		b)	Assume that Alice uses Bob's ElGamal public key (e1=2 and e2=8) to send two messages P = 17 and P′ = 37 using the same random integer r = 9. Eve intercepts the ciphertext and somehow she finds the value of P = 17. Show how Eve can use a known-plaintext attack to find the value of P′.	*CO2*	*PO2*	**7**
		c)	In ElGamal cryptosystem, given the prime p = 31: i. Choose the appropriate values for e_1 and d, then calculate e2. ii. Encrypt the message "HELLO". Use 00 to 25 for encoding. Use different blocks to make P < p. iii. Decrypt the ciphertext to obtain the plaintext. Clearly show all the steps involved in encryption and decryption.	*CO1*	*PO1*	**9**
			OR			
	8	a)	In the elliptic curve E(1, 2) over the GF(11) field: i) Find the equation of the curve.	*CO1*	*PO1*	**10**

			ii) Find at least 5 points on the curve the points on the curve and create a graph representing points on it. iii) Generate public key for Alice using private key = 3 and Generator Point (2,1). iv) Create ciphertext corresponding to the plaintext (4,2) for Alice			
		b)	Consider the following parameters for RSA: e = 7 and p = 19, q=17. a) Write the algorithm b) Encrypt the message block M = 11. c) Compute a private key corresponding to the given public key. d) Perform the decryption of the ciphertext. e) Show if low modulus attack is possible on this message with a suitable example demonstration.	*CO1*	*PO1*	**10**
			UNIT - V			
	9	a)	Using the RSA Digital Signature scheme, let p = 11, q = 19 and d = 23. Calculate the public key e. Then do the following: a. Sign and verify a message with M1 = 12. Calculate the signature S1. b. Sign and verify a message with M2 = 25. Calculate the signature S2. c. Show that if M = M1 × M2 = 300, then S = S1 × S2.	*CO1*	*PO1*	**10**
		b)	Explain the attacks on Digital Signature.	*CO1*	*PO1*	**5**
		c)	Differentiate between conventional signature and a digital signature	*CO2*	*PO2*	**5**
			OR			
	10	a)	In the Diffie-Hellman protocol, what happens if x and y have the same value, that is, Alice and Bob have accidentally chosen the same number? Are R1 and R2 the same? Do the session keys calculated by Alice and Bob have the same value? Use an example to prove your claims.	*CO2*	*PO2*	**5**
		b)	Illustrate with a neat diagram the working of Kerberos protocol. Show and justify the usage of the 3 servers- authentication server, ticket-granting server and a real (data) server in Kerberos.	*CO1*	*PO1*	**10**
		c)	As a security analyst, you are tasked with evaluating the design of the "SecureHash" cryptographic hash function. In your analysis, you come across the Merkle-Damgard scheme, which is used as the underlying construction for this hash function. Considering the importance of the Merkle-Damgard scheme, explain its idea with neat diagram and why it plays a crucial role in the design of a cryptographic hash function like "SecureHash."	*CO2*	*PO2*	**5**

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